

Foreign Keys in Relational Schemas

txt2tex example

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[*RecipeId*, *IngredientId*, *EmployeeId*, *CuisineTag*, *IngredientName*, *RecipeName*, *EmployeeName*, *PositionName*]

Pattern 1: Single-Attribute Primary Key

Each relation has exactly one pk line. txt2tex underlines the attribute in the schema box and emits a PK statement immediately after.

<u>Recipe</u>
<u>recipeId</u> : RecipeId
name : RecipeName
cuisine : CuisineTag

<u>Ingredient</u>
<u>ingredientId</u> : IngredientId
name : IngredientName

Pattern 2: Composite Primary Key

Two pk lines in one schema body. Both attributes are underlined; txt2tex collapses them into one PK set: $\text{PK}(\text{RecipeIngredient}) = \{\text{recipeId}, \text{ingredientId}\}$.

<u>RecipeIngredient</u>
<u>recipeId</u> : RecipeId
<u>ingredientId</u> : IngredientId

Pattern 3: Single-Attribute Foreign Key

A foreign key constraint says every value of the referencing attribute must appear in the referenced relation. No special keyword is needed: write the constraint as a Z predicate using forall/exists or projection with subset.

RecipeIngredient.recipeId must reference an existing Recipe:

$\forall ri : \text{RecipeIngredient} \bullet \exists r : \text{Recipe} \bullet ri.\text{recipeId} = r.\text{recipeId}$

Equivalently, using projection and subset:

$\text{Project}_{\text{recipeId}}(\text{RecipeIngredient}) \subseteq \text{Project}_{\text{recipeId}}(\text{Recipe})$

RecipeIngredient.ingredientId must reference an existing Ingredient:

$\forall ri : \text{RecipeIngredient} \bullet \exists i : \text{Ingredient} \bullet ri.\text{ingredientId} = i.\text{ingredientId}$

Pattern 4: Composite FK and Self-Referential FK

Both FK constraints for *RecipeIngredient* can be stated together in a single zed block. Each predicate is separated by an *also* command in the output.

$$\forall ri : \textit{RecipeIngredient} \bullet \exists r : \textit{Recipe} \bullet ri.\textit{recipeId} = r.\textit{recipeId}$$

$$\forall ri : \textit{RecipeIngredient} \bullet \exists i : \textit{Ingredient} \bullet ri.\textit{ingredientId} = i.\textit{ingredientId}$$

Employee carries a *managerId* that references another row in the same relation. The self-referential FK constraint:

Employee

employeeId : *EmployeeId*

name : *EmployeeName*

position : *PositionName*

managerId : *EmployeeId*

Every *managerId* in *Employee* must match a valid *employeeId* in the same relation:

$$\forall e : \textit{Employee} \bullet \exists m : \textit{Employee} \bullet e.\textit{managerId} = m.\textit{employeeId}$$